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Data Science Capstone Reflection
Carthage College

Capstone Reflection

When I started at Carthage, I thought I would walk away with a basic, well-rounded understanding of data science. What I actually got was so much more than that. Carthage gave me the chance to apply data science and statistics to nearly every area I was curious about, not just one narrow track of it.

That is largely thanks to how I built my concentration. Instead of treating data science as its own isolated subject, I learned it in the context of business, accounting, finance, mathematics and marketing, and then applied it again in computer science and coding. Introduction to Business Analytics showed me the models and processes businesses actually use to make decisions. Marketing Principles connected that same data to real customer behavior. My advanced math classes gave me a deep understanding of numbers, formulas, and how to quickly problem solve. Accounting Fundamentals showed me how it all lands on a financial statement and how those are interpreted. Econometrics and Mathematical Interest Theory stretched the financial math side further. By the time I added in my computer science and coding coursework, I had learned the same underlying skills from four or five different angles, which turned out to matter far more than I expected once I was actually using them at work.

A lot of that came down to my professors. Many of them had come from successful corporate careers before retiring to teach at Carthage, which meant they were not just teaching from a textbook. They had real applications to draw from and genuinely good advice to give, grounded in decades of actually doing the work. That showed up most clearly outside the classroom. My professors went above and beyond, helping me prepare for internship applications and walking me through what interviewers in analytics and business roles were

actually looking for. That kind of mentorship is a big part of why I had the confidence to apply for top internships in the first place, and it is a big part of why I ended up landing two internships of which I am genuinely proud.

The first of those was with Snap-on's Corporate Risk Analysis team, where I built Excel reports and dug through insurance claims and operational data to find patterns leadership had not caught yet. It was the first time I saw a real gap between knowing how to run an analysis and knowing how to make that analysis useful to people who were never going to open the spreadsheet themselves. That internship went well enough that Snap-on brought me back for a second term, and I am currently deep into a much bigger project there, helping build and validate a new reporting system that replaces a set of outdated reports with several the whole risk and dealers' insurance side of the business can rely on, with data checks running all the way back to 2017 and claim transactions going back years earlier.

In between those two terms at Snap-on, I spent a summer at Vertiv on the Data Operations team, where I created the projects of which I am proudest. I built an interactive web app that maps Vertiv's manufacturing footprint, now used by more than 30,000 employees and customers to see sites, business units, and expansion plans in real time. I also built Power BI dashboards across four business units and used AI tools to automate reporting work that used to be done by hand. Vertiv was also where I presented analysis to leadership for the first time, which forced me to translate a dashboard into a decision instead of just a chart.

Outside of internships, I kept pushing in directions that were not required by my major at all: building tableau dashboards, taking online courses on Power BI, practicing making models in Python, and building an AI automation agent just to see what I could get it to do on its own. I played lacrosse and stayed involved in clubs on campus too, which taught me a similar kind of consistency to the one my coursework did, showing up and doing the

work even when the results are not immediate. I am also starting work next semester helping build a new Artificial Intelligence and Machine Learning course at Carthage, which feels like its own kind of full circle, given how much I underestimated this field when I started.

If I am honest, the biggest thing I learned at Carthage was not a single tool or technique. It was that data science is not one narrow subject, it is a lens I can bring to almost anything, whether that is a balance sheet, a customer campaign, a line of code, or a conversation with an executive. Four years ago I expected to leave with a basic understanding. Instead I am leaving with a concentration I designed myself, two internships at companies I am proud to have worked for, and professors who treated my career like it mattered as much as my grade did. I am looking forward to finding out where that takes me next.